Local robustness of commuting-reflection encoding near the three-qubit W state

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Abstract

A pure state is locally encodable when product unitaries generate a complete orthonormal basis from it. We consider the restricted class of coordinated encoders with Hermitian involutory local factors and pairwise commuting global operators. Starting from the known triangle-graph encoding of the three-qubit W state, we prove that every sufficiently small arbitrary complex pure-state perturbation remains exactly encodable with the same graph. Although the seven moment constraints have a rank-six control Jacobian at the reference encoding, elimination of six regular constraints leaves a nonzero cubic response. A uniform sign argument gives local existence and an anchored control-displacement bound of order the cube root of the state displacement. An explicit normalized state path admits cube-root continuation but no continuously differentiable encoder branch through the reference control. The neighborhood radius and bound constants are existential. The result does not resolve universal three-qubit encodability.

1 Encoding problem and relation to earlier results

A complete measurement basis can encode different classical labels while retaining the entanglement of a single reference state. Pimpel, Renner, and Tavakoli studied bases generated by general product unitaries, including numerical evidence for three-qubit states and constructions for real states [1]. Pauwels and Gühne characterized the more restrictive commuting-reflection class by flatness in a graph-state basis up to local unitaries, and constructed an exact encoding of the W state [2, Theorem 6 and Proposition 7]. Their discussion conjectures that every three-qubit pure state admits such an encoding.

We prove a local robustness result for that restricted class: the published W encoding extends to a full neighborhood in the space of complex pure-state rays. The control derivative is singular, so a regular seven-equation implicit-function argument cannot establish this extension. A cubic response in the remaining constraint supplies the missing mechanism. The earlier real-state construction permits general local encoders and does not supply this neighborhood in the commuting-reflection class. We are not aware of a prior proof of this precise local statement in the primary literature checked; this comparison does not establish exhaustive priority, particularly with respect to unpublished work.

Let $|\psi\rangle \in (\mathbb{C}^2)^{\otimes 3}$ be normalized. An encoding consists of eight product unitaries U_α such that each local factor is Hermitian and squares to the identity, the global operators commute, and

$$\langle\psi|U_\alpha^\dagger U_\beta|\psi\rangle = \delta_{\alpha\beta}. \tag{1}$$

Each party may use the full common label α. Thus coordination of the classical controls is allowed. No entangling encoder is applied. Decoding is a joint projective measurement; local distinguishability is not assumed.

2 Triangle moments and reference control

Use standard Pauli matrices X, Y, Z, with $Y = iXZ$, and define the triangle stabilizer generators

$$K_1 = XZZ, \qquad K_2 = ZXZ, \qquad K_3 = ZZX. \tag{2}$$

For $\alpha \in \mathbb{F}_2^3$, write $K^\alpha = \prod_j K_j^{\alpha_j}$. Their joint eigenbasis is

$$|G_\beta\rangle = Z^\beta CZ_{12} CZ_{13} CZ_{23} |+\rangle^{\otimes 3}. \tag{3}$$

For a product $V \in SU(2)^{\otimes 3}$, set $z = V\psi$. The probabilities in this basis obey the Fourier relation

$$\left|\langle G_\beta | z\rangle\right|^2 = \frac{1}{8} \sum_{\alpha \in \mathbb{F}_2^3} (-1)^{\alpha\cdot\beta} \langle z | K^\alpha | z\rangle. \tag{4}$$

Consequently the seven nonidentity moments vanish exactly if and only if all eight probabilities equal 1/8. In that case $U_\alpha = V^\dagger K^\alpha V$ satisfies Eq. (1), because the Gram entries are the moments of $K^{\alpha+\beta}$. Products of the commuting K_j are signed Pauli tensors. Their overall real sign can be assigned to one local factor, so every U_α has legal local Hermitian involutory factors. The graph basis here is a mathematical representation of the constraints.

The known W solution [2, Proposition 7] is

$$|W_3\rangle = \frac{|001\rangle + |010\rangle + |100\rangle}{\sqrt{3}}, \qquad V_0 = q^{\otimes 3}, \qquad q = i(bY + aZ), \tag{5}$$

where $a, b > 0$ and

$$a^2 = \frac{3+\sqrt{3}}{6}, \quad b^2 = \frac{3-\sqrt{3}}{6}, \quad ab = \frac{1}{\sqrt{6}}. \tag{6}$$

All seven moments of $z_0 = V_0 W_3$ vanish. Local controls near V_0 are parametrized by

$$V(\theta) = \exp\left[-\frac{i}{2}\sum_{j=1}^{3} (\theta_{jX} X_j + \theta_{jY} Y_j + \theta_{jZ} Z_j)\right] V_0, \tag{7}$$

with real $\theta \in \mathbb{R}^9$ and qubit-major ordering $(1X, 1Y, 1Z, 2X, \ldots, 3Z)$. The exponential factorizes across qubits. States, moments, and controls are dimensionless. These are local coordinate expansions of exact unitaries, not a finite-order physical approximation.

3 Local existence despite the singular derivative

Theorem 1. *There exist $\epsilon > 0$ and $C < \infty$ such that every state ray represented by*

$$\psi(\eta) = \frac{W_3 + \eta}{\sqrt{1 + \|\eta\|^2}}, \qquad \langle W_3 | \eta\rangle = 0, \qquad \|\eta\| < \epsilon, \tag{8}$$

admits a commuting-reflection encoding using the fixed triangle graph, with a control in Eq. (7) satisfying

$$\|\theta\| \le C \|\eta\|^{1/3}. \tag{9}$$

The constants are existential, and the choice is anchored at $\theta = 0$ for $\eta = 0$. No continuous selection over all nearby states is asserted.

The phase in Eq. (8) is fixed by positive real overlap with W_3. The orthogonal η contains fourteen real coordinates and covers arbitrary complex perturbations of the ray.

3.1 Six regular constraints

Let F be the six expectations of

$$S = (K_1, K_2, K_3, K_1K_2, K_1K_3, K_2K_3) \tag{10}$$

in $V(\theta)\psi(\eta)$, and introduce the missing observable

$$O = K_1K_2K_3 - 2(K_1 + K_2 + K_3). \tag{11}$$

The equations $F = 0$ and $\langle O\rangle = 0$ are equivalent to all seven moments vanishing. At the reference point the full control derivative of $\langle O\rangle$ is zero. The derivative of a moment T is

$$\partial_{\theta_{jk}}\langle T\rangle = \frac{i}{2}\langle z_0|[\sigma_{jk}, T]|z_0\rangle. \tag{12}$$

Restrict to the six columns $(Z_1, Z_2, Z_3, X_1, X_2, X_3)$. Direct evaluation yields

$$J_B = \begin{pmatrix} dI_3 & 0 \\ 0 & eA \end{pmatrix}, \qquad d = \frac{2ab}{3}, \quad e = \frac{\sqrt{2}}{3}, \quad A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}. \tag{13}$$

It has determinant $-16\sqrt{3}/6561 \neq 0$. Together with the zero derivative of O, this shows that the seven-constraint Jacobian has rank exactly six.

Three kernel controls v_h, $h = 1, 2, 3$, are specified by

$$(v_h)_{jX} = 0, \qquad (v_h)_{jY} = \sqrt{2}\delta_{jh}, \qquad (v_h)_{jZ} = 2 - 5\delta_{jh}. \tag{14}$$

Let N have these columns, and let B insert the six Z, X coordinates in the ordering of Eq. (13). The nine columns of (N, B) are independent because the Y components of N are invertible. With $\theta = Nu + By$, the real analytic implicit-function theorem produces

$$y = \mathcal{Y}(u, \eta), \qquad F(Nu + B\mathcal{Y}(u, \eta), \eta) = 0, \tag{15}$$

near the origin, with $\mathcal{Y}(0, 0) = 0$ and $D_u\mathcal{Y}(0, 0) = 0$.

3.2 The decisive cubic coefficient

It suffices to set $u = se_1$ and study the remaining scalar

$$r(s, \eta) = \langle V(\theta(s, \eta))\psi(\eta)|O|V(\theta(s, \eta))\psi(\eta)\rangle, \quad \theta(s, \eta) = sv_1 + B\mathcal{Y}(se_1, \eta). \tag{16}$$

At $\eta = 0$, the second-order control correction is

$$\theta(s, 0) = sv_1 + s^2c + O(s^3), \qquad c = \frac{1}{\sqrt{2}}(13, 0, 0, -4, 0, 0, -4, 0, 0). \tag{17}$$

The six regular moments and the missing moment vanish through order two along this curve. Define $H = \frac{1}{2}\sum(v_1)_{jk}\sigma_{jk}$ and $L = \frac{1}{2}\sum c_{jk}\sigma_{jk}$. With expectation in z_0, the directly checkable identities are

$$\begin{aligned} &\langle S\rangle_\ell = 0, && i\langle[H, S_\ell]\rangle = 0, && i\langle[L, S_\ell]\rangle - \tfrac{1}{2}\langle[H, [H, S_\ell]]\rangle = 0, \\ &\langle O\rangle = 0, && i\langle[H, O]\rangle = 0, && i\langle[L, O]\rangle - \tfrac{1}{2}\langle[H, [H, O]]\rangle = 0. \end{aligned} \tag{18}$$

The cubic coefficient of the missing moment is

$$c_0 = -\frac{i}{6}\langle [H,[H,[H,O]]]\rangle - \frac{1}{2}\langle [H,[L,O]] + [L,[H,O]]\rangle = -\frac{80ab}{3} \neq 0. \tag{19}$$

An unknown third-order complementary control contributes only a first derivative of O, which vanishes for every control. Hence it cannot change Eq. (19), and the exact reduced function satisfies

$$r(s,0) = c_0 s^3 + O(s^4). \tag{20}$$

These identities can be evaluated using only 8×8 matrices and rational radical arithmetic. For example, conjugating operators back into the original W basis uses

$$q^\dagger X q = -X, \quad q^\dagger Y q = -\frac{Y}{\sqrt{3}} + \sqrt{\frac{2}{3}}Z, \quad q^\dagger Z q = \sqrt{\frac{2}{3}}Y + \frac{Z}{\sqrt{3}}. \tag{21}$$

For any resulting matrix T, its W expectation is $\frac{1}{3}\sum_{m,n\in\{1,2,4\}} T_{mn}$ in zero-based computational indices. These formulas provide a short exact reproduction route for Eqs. (13)–(19).

3.3 Uniform sign argument

Choose $\rho > 0$ within the analytic neighborhood so that

$$|r(s,0) - c_0 s^3| \leq \tfrac{1}{2}|c_0||s|^3 \qquad (|s| \leq \rho). \tag{22}$$

On a smaller compact product neighborhood, smooth state dependence gives a finite $M \geq 1$ such that

$$|r(s,\eta) - r(s,0)| \leq M\|\eta\|. \tag{23}$$

Put $d_\eta = \|\eta\| > 0$ and $A_0 = (4M/|c_0|)^{1/3}$. For sufficiently small d_η, the two values $s_\pm = \pm A_0 d_\eta^{1/3}$ lie within $[-\rho,\rho]$. The unperturbed residuals have opposite signs and magnitudes at least $2Md_\eta$. The perturbed residuals therefore retain opposite signs. The intermediate-value theorem gives an exact zero s_* between them. At $\eta = 0$ take $s_* = 0$.

All six eliminated equations hold exactly throughout this interval. Analyticity and $D_u\mathcal{Y}(0,0) = 0$ give

$$\|\mathcal{Y}(se_1,\eta)\| \leq C_1(s^2 + \|\eta\|). \tag{24}$$

Thus $\theta(s_*,\eta)$ satisfies Eq. (9) after shrinking the neighborhood. Equations (4) and (1) complete the proof of Theorem 1. The Taylor expansion supplies a sign with a bounded analytic remainder; the encoded states solve the exact equations.

4 A state path with cube-root compensation

The singular response can be isolated on a legal state path. Set

$$\delta = V_0^\dagger O V_0 W_3, \qquad \psi(t) = \frac{W_3 + t\delta}{\sqrt{1+13t^2}}, \qquad t \in \mathbb{R}. \tag{25}$$

Direct evaluation gives $\langle W_3|\delta\rangle = 0$, $\|\delta\|^2 = 13$, and

$$\frac{d}{dt}\langle V_0\psi(t)|O|V_0\psi(t)\rangle\Big|_{t=0} = 26. \tag{26}$$

An explicit expression is

$$\delta = \frac{1}{\sqrt{3}}\left(2|000\rangle + |011\rangle + |101\rangle + |110\rangle - 8i\sqrt{3}ab|111\rangle\right). \tag{27}$$

After eliminating the six regular moments, the scalar equation is

$$r_{\mathrm{path}}(s,t) = 26t + c_0 s^3 + O(s^4 + |t||s| + t^2). \tag{28}$$

The derivative in t is nonzero, so the analytic implicit-function theorem yields an exact solution curve

$$t = T(s) = \frac{40ab}{39}s^3 + O(s^4), \qquad \theta = sv_1 + O(s^2). \tag{29}$$

Since $T'(s) = 3(40ab/39)s^2 + O(s^3) > 0$ for small nonzero s, it has a two-sided local inverse. The resulting control branch satisfies

$$\theta(t) = v_1\left(\frac{39t}{40ab}\right)^{1/3} + O(|t|^{2/3}), \tag{30}$$

where the real cube root is used. In contrast, any C^1 control path through V_0 contributes zero first-order variation of the missing moment. It cannot cancel the nonzero state derivative 26. There is therefore no C^1 encoder branch through this particular reference control on this state path. This does not exclude other reference encoders or graphs, and it does not establish optimality of the exponent over all encoder choices.

5 Physical meaning, reproducibility, and limitations

The result preserves eight perfectly distinguishable encoded messages for every sufficiently small complex pure-state perturbation of W, using one fixed connected graph and coordinated product reflections. It establishes exact local robustness without assuming real amplitudes. It also explains how a singular control Jacobian can coexist with local existence: a cubic control response compensates a linear state perturbation.

No numerical neighborhood radius, practical globally convergent solver, mixed-state noise guarantee, ancillary resource, or LOCC decoding claim is supplied. The control bound is anchored at W and is not a pairwise Hölder-continuity statement for a global encoder selection. The original universal three-qubit conjecture remains unresolved away from the proved neighborhood.

The accompanying reproducibility instructions identify the exact arithmetic code, input conventions, and outputs for the reference moments, regular derivative, and cubic jet. The decisive coefficient is obtained from exact nested commutators and independently checked in the original W basis. Finite Taylor orders are used only to prove a sign property of analytic functions; no fitted numerical residual is treated as an exact encoding.

AI use in the research methods. OpenAI Codex tools assisted with the local differential analysis, cubic reduction, analytic existence argument, exact arithmetic code, literature comparison, and computational checks. Separate AI research and checking contexts were used; these checks are not human scientific verification or external peer review. The precise model versions used in earlier research phases remain to be confirmed from the retained run records. Human review of the derivation, code, and interpretation is pending for this local draft.

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References

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